Formalizing Bombieri Vinogradov

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Chapter 1

Bombieri–Vinogradov

1.1 Statement

The main result is the Bombieri–Vinogradov theorem, which gives a strong estimate for primes in arithmetic progressions on average over moduli.

Theorem 1. *For each fixed $A \geq 0$,*

$$\sum_{q \leq Q} \max_{y \leq x} \max_{a \in (\mathbb{Z}/q\mathbb{Z})^*} \left| \psi(y; q, a) - \frac{y}{\varphi(q)} \right| \ll_A \frac{x}{(\log x)^A}$$

uniformly for all $x \geq 2$ and $1 \leq Q \leq \frac{\sqrt{x}}{(\log x)^{A+3}}$.

Proof. Apply **??** and absorb the error terms. □

Theorem 2. *For each fixed $A \geq 0$ we have*

$$\sum_{q \leq Q} \max_{y \leq x} \max_{a \in (\mathbb{Z}/q\mathbb{Z})^*} |\Delta_{1_P}(y; q, a)| \ll_A \frac{x}{(\log x)^{A+1}}$$

uniformly for $x \geq 2$ and $1 \leq Q \leq \sqrt{x}/(\log(x))^{A+3}$

Proof. Apply **??** and **??**; the error terms $\sum_{q \leq Q} \sqrt{x}/\varphi(q) \cdot (1/\log x)$ are absorbed by $x/(\log x)^{A+1}$. □

1.2 Assumptions

We take Siegel–Walfisz and the large sieve inequality as axioms.

Definition 3. Let $A, C > 0$. If $1 \leq q \leq (\log x)^C$ and $a \in (\mathbb{Z}/q\mathbb{Z})^*$, then

$$\psi(x; q, a) = \frac{x}{\varphi(q)} + O_{A,C}\left(\frac{x}{(\log x)^A}\right).$$

Definition 4. Let $Q \geq 1$, $H \in \mathbb{Z}$, $N \in \mathbb{Z}_{\geq 1}$ and $c = (c_{H+1}, \ldots, c_{H+N}) \in \mathbb{C}^N$ We then have

$$\sum_{q \leq Q} \sideset{}{^*}\sum_{\chi \pmod q} \frac{q}{\varphi(q)} \left| \sum_{H < n \leq H+N} c_n \chi(n) \right|^2 \ll (N + Q^2) \|\vec{c}\|_2^2,$$

1.3 Definitions

Definition 5. $\mathrm{li}(x) = \int_2^x \frac{\mathrm{d}t}{\log t}$

Definition 6.

$$\Delta_f(x; q, a) := \sum_{n \le x,\ n \equiv a \pmod q} f(n) \ - \frac{1}{\varphi(q)} \sum_{n \le x, (n,q)=1} f(n)$$

for $x \ge 1$, $q \in \mathbb{N}$

Theorem 7.

$$\Delta_f(y; q, a) = \frac{1}{\varphi(q)} \sum_{\chi \pmod q, \chi \neq \chi_0} \bar{\chi}(a) \sum_{n \le y} f(n)\chi(n)$$

Proof. □

Definition 8. For $f : \mathbb{N} \to \mathbb{R}$ and $r : \mathbb{N}$ we use f_r to denote $n \mapsto f(n)1_{(n,r)=1}$

1.4 Reduction to Δ_Λ

Bombieri–Vinogradov for ψ is reduced to a bound for Δ_Λ via the following steps.

Theorem 9.

$$\Delta_{1_P}(x; q, a) = \pi(x; q, a) - \frac{1}{\varphi(q)} \sum_{p \le x, p \nmid q} 1$$

Proof. □

Theorem 10.

$$\sum_{p \le x, p \nmid q} 1 = li(x) + O(xe^{-c\sqrt{\log x}} + \log q)$$

Proof. □

Theorem 11. *For $x \ge 2$, $q \in \mathbb{N}$ and $a \in (\mathbb{Z}/q\mathbb{Z})^*$ we have*

$$\max_{y \le x} |\Delta_{1_P}(y; q, a)| \ll \frac{1}{\log x} \left(\max_{\sqrt{x} \le y \le x} |\Delta_\Lambda(y; q, a)| + \sqrt{x} \right)$$

Proof. Uses combinatorics, an estimate on $\sum_{p \le x} 1/\log p$, and partial summation. See Koukoulopoulos p. 279. □

Theorem 12. *For each fixed $A \ge 0$ we have*

$$\sum_{q \le Q} \max_{y \le x} \max_{a \in (\mathbb{Z}/q\mathbb{Z})^*} |\Delta_{1_P}(y; q, a)| \ll_A \frac{x}{(\log x)^{A+1}}$$

uniformly for $x \ge 2$ and $1 \le Q \le \sqrt{x}/(\log(x))^{A+3}$

Proof. Apply **??** and **??**; the error terms $\sum_{q \le Q} \sqrt{x}/\varphi(q) \cdot (1/\log x)$ are absorbed by $x/(\log x)^{A+1}$. □

Theorem 13. *For each fixed $A \geq 0$ we have*

$$\sum_{q \leq Q} \max_{\sqrt{x} \leq y \leq x} \max_{a \in (\mathbb{Z}/q\mathbb{Z})^*} |\Delta_\Lambda(y; q, a)| \ll_A \frac{x}{(\log x)^A}$$

uniformly for $x \geq 2$ and $1 \leq Q \leq \sqrt{x}/(\log(x))^{A+3}$

Proof. Follows from **??** and the triangle inequality, combining the bounds **??**, **??**, and **??**. □

1.5 Vaughan's Identity

The von Mangoldt function decomposes as $\Lambda = \Lambda^\sharp + \Lambda^\flat + \Lambda_{\leq U}$, where $UV \leq \sqrt{x}$ and $U, V \geq e^{\sqrt{\log x}}$.

Definition 14. $\Lambda^\sharp = \mu_{\leq V} * \log - (\Lambda_{\leq U} * \mu_{\leq V}) * 1$

Definition 15. $\Lambda^\flat = (\Lambda_{>U} * 1) * \mu_{>V}$

Definition 16. $\Lambda_{\leq U} = 1_U \cdot \Lambda$

Theorem 17. *Decompose $\Lambda = \Lambda^\sharp + \Lambda^\flat + \Lambda_{\leq U}$*

Proof. □

1.6 Small Primes ($\Lambda_{\leq U}$)

The contribution of the small part $\Lambda_{\leq U}$ is bounded trivially.

Lemma 18.

$$\sum_{n \leq U} \Lambda(n) \ll U$$

Proof. □

Lemma 19. *For $y, U > 0$, $q \in \mathbb{N}$ and $a \in \mathbb{Z}/q\mathbb{Z}$,*

$$|\Delta_{\Lambda_{\leq U}}(y;\, q,\, a)| \ll U$$

Proof. □

Theorem 20. *For each fixed $A \geq 0$, $x \geq 2$ and $1 \leq Q \leq \sqrt{x}/(\log x)^{A+3}$,*

$$\sum_{q \leq Q} \max_{\sqrt{x} \leq y \leq x} \max_{a \in (\mathbb{Z}/q\mathbb{Z})^*} |\Delta_{\Lambda_{\leq U}}(y;\, q,\, a)| \leq Q\sqrt{x} \ll_A \frac{x}{(\log x)^{A+2}}$$

Proof. □

1.7 Type I Sums ($\Lambda^\sharp$)

The sharp part $\Lambda^\sharp$ is a type I sum, bounded via Abel summation.

Lemma 21. *If f is an arithmetic function supported on $[1, y]$ then*

$$\Delta_{f*g}(x;\, q,\, a) = \sum_{\substack{k \le y \\ (k,q)=1}} f(k)\, \Delta_g\Big(\frac{x}{k};\, q,\, a\bar{k}\Big)$$

Proof. □

Lemma 22. *For $x \ge 1$, $q \in \mathbb{N}$ and $a \in (\mathbb{Z}/q\mathbb{Z})^*$,*

$$|\Delta_1(x;\, q,\, a)| \le 1$$

Proof. Carefully consider length q intervals. Alternatively, write

$$\Delta_1(t;\, q,\, a) = \frac{1}{\varphi(q)} \sum_{a' \in (\mathbb{Z}/q\mathbb{Z})^*} \left(\sum_{\substack{n \le t \\ n \equiv a \pmod q}} 1 - \sum_{\substack{n \le t \\ n \equiv a' \pmod q}} 1 \right)$$

and note each inner difference is bounded by 1 in absolute value. □

Lemma 23. *If g is continuously differentiable on $[1, x]$ then*

$$\Delta_g(x;\, q,\, a) = \Delta_1(x;\, q,\, a)\, g(x) - \int_1^x \Delta_1(t;\, q,\, a)\, g'(t)\, \mathrm{d}t$$

Proof. By Abel summation. □

Lemma 24. *If g is continuously differentiable and monotone on $[1, y]$ with $g(0) = 0$, then for all $t \ge 1$ and $a \in (\mathbb{Z}/q\mathbb{Z})^*$,*

$$|\Delta_g(x;\, q,\, a)| \le 2g(t)$$

Proof. □

Theorem 25. *Let $v \ge 0$ and let f be an arithmetic function supported on $[1, y]$. For $x \ge 2$, $q \in \mathbb{N}$ and $a \in (\mathbb{Z}/q\mathbb{Z})^*$,*

$$|\Delta_{f*\log^v}(x;\, q,\, a)| \le 2(\log x)^v \sum_{k \le y} |f(k)|$$

Proof. Straightforward application of the previous lemmas. □

Theorem 26. *For $U, V \ge 1$, $x \ge 2$, $q \in \mathbb{N}$, $r \le x$ and $a \in (\mathbb{Z}/q\mathbb{Z})^*$,*

$$\max_{y \le x} \max_{a \in (\mathbb{Z}/q\mathbb{Z})^*} |\Delta_{\Lambda_r^\sharp}(y;\, q,\, a)| \ll UV \log x$$

Proof. Apply **??** twice: once with $f = \mu_{\le V,\, r}$, $v = 0$, and once with $f = \Lambda_{\le U} * \mu_{\le V,\, r}$, $v = 0$, $y = UV$. □

Theorem 27. *For each fixed $A \ge 0$, $x \ge 2$ and $1 \le Q \le \sqrt{x}/(\log x)^{A+3}$,*

$$\sum_{q \le Q} \max_{\sqrt{x} \le y \le x} \max_{a \in (\mathbb{Z}/q\mathbb{Z})^*} |\Delta_{\Lambda^\sharp}(y;\, q,\, a)| \ll_A \frac{x}{(\log x)^A}$$

Proof. □

1.8 Type II Sums ($\Lambda^\flat$)

1.8.1 Definitions

Definition 28.

$$S_r(y,\xi) := \left|\sum_{n\le y} \Lambda_r^\flat(n)\,\xi(n)\right|$$

Definition 29.

$$T_r(x,Q) := \sum_{(\log x)^C < d \le Q/r} \frac{1}{\varphi(d)} \sideset{}{^*}\sum_{\xi \pmod d} \max_{\sqrt{x}\le y\le x} S_r(y,\xi)$$

1.8.2 Reduction to character sums

Lemma 30. *This is a standard result. Let f be a function from Dirichlet characters. Then*

$$\sum_{\substack{\chi \pmod q \\ \chi\neq\chi_0}} f(\chi) = \sum_{d|q} \sideset{}{^*}\sum_{\xi \pmod d} f(1_{(n,q)=1}\xi)$$

Proof. □

Lemma 31. *Let f be an arithmetic function. For $r \le x$, $q > 1$ and $(a,q) = 1$,*

$$\sideset{}{^*}\sum_{\xi \pmod q} \bar{\xi}(a) \sum_{n\le y} \xi(n) f_r(n) = \sum_{d|q} \mu(q/d)\,\varphi(d)\,\Delta_{f_{rq}}(y;\,d,\,a)$$

Proof. Fix $P \in \mathbb{N}$ with $q \mid P$. Define F_P and G_P on divisors of P by

$$F_P(q) := \sum_{\chi\neq\chi_0 \pmod q} \bar{\chi}(a) \sum_{n\le y} \chi(n) f_{rP}(n) = \Delta_{f_{rP}}(y;\,q,\,a),$$
$$G_P(d) := \sideset{}{^*}\sum_{\xi \pmod d} \bar{\xi}(a) \sum_{n\le y} \xi(n) f_{rP}(n) \quad (d>1), \quad G_P(1) = 0.$$

Since every non-principal character mod q factors through a unique primitive character, $F_P(q) = \sum_{d|q} G_P(d)$. By Möbius inversion,

$$G_P(q) = \sum_{d|q} \mu(q/d)\,F_P(d) = \sum_{d|q} \mu(q/d)\,\Delta_{f_{rP}}(y;\,d,\,a).$$

Set $P = q$ to conclude. □

Lemma 32.

$$|\Delta_{\Lambda^\flat}(y;\,q,\,a)| \le \frac{1}{\varphi(q)} \left| \sum_{\substack{d|q \\ 1<d\le(\log x)^C}} \sum_{s|d} \mu(d/s)\,\varphi(s)\,\Delta_{\Lambda_q^\flat}(y;\,s,\,a) \right| + \frac{1}{\varphi(q)} \sum_{\substack{d|q \\ d>(\log x)^C}} \sideset{}{^*}\sum_{\xi \pmod d} S_{q/d}(y,\xi)$$

Proof. □

Theorem 33.

$$\frac{1}{\varphi(q)}\left|\sum_{\substack{d\mid q\\ 1<d\le(\log x)^C}}\sum_{s\mid d}\mu(d/s)\,\varphi(s)\,\Delta_{\Lambda_q^\flat}(y;\,s,\,a)\right| \ll_{A,C} \frac{x}{\varphi(q)\,(\log x)^{A+1}}$$

Proof. Push the absolute values inside, then

$$\begin{aligned}\sum_{\substack{d\mid q\\ d\le(\log x)^C}}\sum_{s\mid d}\varphi(s)\,\left|\Delta_{\Lambda_q^\flat}(y;\,s,\,a)\right| &\ll_{A,C} \sum_{d\le(\log x)^C}\left(\sum_{s\mid d}\varphi(s)\right)\frac{x}{(\log x)^{A+2C+1}}\\ &\ll \frac{x}{(\log x)^{A+2C+1}}\sum_{d\le(\log x)^C} d\\ &\ll \frac{x}{(\log x)^{A+1}}.\end{aligned}$$

□

Theorem 34.

$$\sum_{q\le Q}\max_{\substack{\sqrt{x}\le y\le x\\ a\in(\mathbb{Z}/q\mathbb{Z})^*}}|\Delta_{\Lambda^\flat}(y;\,q,\,a)| \le \sum_{r\le Q}\frac{T_r(x,Q)}{\varphi(r)} + O\left(\frac{x}{(\log x)^A}\right)$$

Proof. Sum the error from **??** over $q\le Q$ using $\sum_{n\le x}1/\varphi(n)\ll\log x$, then regroup the main sum by $r=q/d$. □

1.8.3 Large sieve estimates

Theorem 35. *Let f and g be arithmetic functions supported on $[1,M]$ and $[1,N]$ respectively. For $x,Q\ge 1$,*

$$\sum_{q\le Q}\sum_{\chi}{}^{*}_{(\mathrm{mod}\ q)}\frac{q}{\varphi(q)}\max_{y\le x}\left|\sum_{n\le y}(f*g)(n)\chi(n)\right| \ll \left(\sqrt{MN}+\sqrt{M}\,Q+\sqrt{N}\,Q+Q^2\right)(\log x)\,\|f\|_2\,\|g\|_2$$

Proof. Uses Cauchy–Schwarz and the large sieve inequality (**??**). The smooth truncation uses a version of Perron's integral formula. □

Lemma 36.

$$\Lambda^\flat(n) = \sum_{U<2^j\le 2x/V}(f_j*g_j)(n)\quad \textit{for } n\le x,$$

where $f_j(k)=(\Lambda_{>U}*1)(k)\,1_{2^{j-1}<k\le 2^j}$ and $g_j(\ell)=\mu(\ell)\,1_{V<\ell\le x/2^{j-1}}$.

Proof. □

Theorem 37. *For $x,Q\ge 2$, $U,V\in[1,x]$ and $r\in\mathbb{N}$,*

$$\sum_{q\le Q}\sum_{\chi}{}^{*}_{(\mathrm{mod}\ q)}\frac{q}{\varphi(q)}\,S_r(y,\chi) \ll \left(x+\frac{Qx}{\sqrt{U}}+\frac{Qx}{\sqrt{V}}+Q^2\sqrt{x}\right)(\log x)^3$$

Proof. Apply the dyadic decomposition **??** (restricted to integers coprime to r) and apply **??** to each dyadic piece. When summing over j note $U \le 2^j$, so $\sum_{U \le 2^j} 2^{-j/2} \ll 1/\sqrt{U}$. □

Theorem 38.

$$T_r(x, Q) \ll \frac{x}{(\log x)^{C-3}} + \frac{x(\log x)^4}{\sqrt{U}} + \frac{x(\log x)^4}{\sqrt{V}} + \frac{Q\sqrt{x}\,(\log x)^3}{r}$$

Proof. Divide the sum defining T_r into dyadic intervals in d and apply **??**. □

Theorem 39. *For each fixed $A \ge 0$, $x \ge 2$ and $1 \le Q \le \sqrt{x}/(\log x)^{A+3}$,*

$$\sum_{q \le Q} \max_{\sqrt{x} \le y \le x} \max_{a \in (\mathbb{Z}/q\mathbb{Z})^*} |\Delta_{\Lambda^\flat}(y;\, q,\, a)| \ll_A \frac{x}{(\log x)^A}$$

Proof. Plug the bound from **??** into **??**, then choose $U = V = e^{\sqrt{\log x}}$ and $C = A + 4$. □